

1. PROBLEM STATEMENT

Weightlifting is a widely practiced form of exercise, but it presents challenges for both beginners and experienced participants. Many individuals struggle with maintaining proper form, understanding gym equipment, and selecting appropriate weights or workout plans. These challenges often result in injuries, frustration, and decreased motivation for those seeking safe and effective training.

1.1 Need Statement

Improper form during weightlifting contributes to injuries, with approximately 970,000 weightlifting-related injuries reported in the United States between 1990 and 2007 [1]. Many beginners find gyms intimidating because of the complexity of the equipment and the absence of clear guidance [2]. Selecting appropriate weights and exercises often requires significant research, which can overwhelm new users [3]. Current fitness tools, including mobile applications and basic wearables, provide data such as steps, calories, or repetitions, but they lack the ability to deliver real-time feedback on form [4]. This gap leaves individuals at risk of injury and limits their ability to train effectively without professional supervision.

1.2 Objective Statement

The objective of this project is to design a portable system (all detachable sensors located in a tote-around case) that improves weightlifting safety and effectiveness. This comprehensive system - dubbed the *Repetition. Efficiency. Precision (R.E.P.) Tracker* - utilizes advanced accelerometer tracking, and easy-to-use heart rate, oxygen, and temperature sensors to sense the user's current performative state (e.g. weightlifting form and workout intensity); the system's user-feedback, user-interface, and a camera recorder provide quality assessments through vocal (text-to-speech programming) and visual (recordable camera feed) outputs to better recommend exercise and body adjustments to the user. These guided outputs and metrics are based on the data processed from the accelerometer and the aforementioned sensors. To summarize, the R.E.P Tracker system analyzes body motion, counts repetitions and sets, and communicates feedback in real time to guide proper technique and monitor long-term progress so that the user can have a stronger, safer, and healthier fitness journey.

1.3 Background and Related Work

Motion tracking and camera-based monitoring are widely used in fitness technology. Consumer wearables such as Fitbit track steps and calories but do not analyze movement patterns in detail [4]. High-precision systems such as Vicon deliver accurate motion capture for research and clinical applications, but they are expensive and impractical for general gym use [5]. Smart home gym products such as Mirror and Tempo Studio provide form feedback through large interactive screens, but their high cost and subscription requirements reduce accessibility [6]. Existing patents on motion-tracking technologies demonstrate general approaches to athletic monitoring but do not directly address affordable, portable weightlifting feedback systems [7]. This project focuses on bridging that gap by combining portability, low cost, and real-time form analysis in a compact solution.

2. Design Requirement Specifications

2.1 Marketing Requirements

- Designed for beginner to intermediate weightlifters with less gym experience.

- Must be affordable and portable, with a target cost of under \$500 for the initial product.
- Setup should be simple, taking less than 5 minutes.
- Must provide real-time feedback to help prevent injuries and build user confidence.
- Should support long-term progress tracking and allow users to customize workout plans.

2.2 Engineering Requirements

- Motion sensors must track reps, sets, and joint angles with an accuracy of $\pm 2^\circ$.
- Camera module must capture 1080p video at 30 FPS for form analysis.
- Charging case should last 2–3 days on a 2000mAh battery.
- Wireless BLE communication must keep latency under 100 ms.
- System must work in temperatures from 0°C to 40°C and up to 85% humidity.

2.3 Constraints

- The project budget is limited to \$700, including both hardware and development.
- The full system (case and components) must weigh under 5 kg to remain portable.
- The design must work in gyms with limited space and variable lighting.
- The project must be completed by end of April

3. REFERENCES

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4. ACKNOWLEDGEMENT

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